SOFTWARE CONSTRUCTION · LECTURE 5

User Interface Design

What & Why?



What is UI Design?

> *The design of user interfaces for machines and software, such as computers, home appliances, mobile devices, and other electronic devices, with the focus on maximizing usability and the user experience. The aim is to make the user's interaction as simple and efficient as possible, in terms of accomplishing user goals.*
>
> **— Wikipedia**

In Simple Words

The design of user interfaces for different software or machines with a clear purpose.

Make a better experience for users when navigating through your app, platform or website.

What UI Actually Covers

- The visual elements of a product or digital experience.
- A translation of a brand's vision into a product or a software.
- Common UIs are found on: the phone, computer, or tablet; at the ATM; and in your car.

NEXT TOPIC

User Experience Design

What & Why?



What is UX Design?

> *User experience design (UXD, UED, or XD) is the process of enhancing user satisfaction with a product by improving the usability, accessibility, and pleasure provided in the interaction with the product. It encompasses traditional human–computer interaction (HCI) design and extends it by addressing all aspects of a product or service as perceived by users.*
>
> **— Wikipedia**

In Simple Words

UX Design is studying user behaviour and understanding user motivations with the goal of designing better digital experiences.

Make a better experience for users by understanding their behaviour and how they interact with a product or a software.

UX is NOT UI

Seriously!

Two Different Disciplines

Wireframes

UX

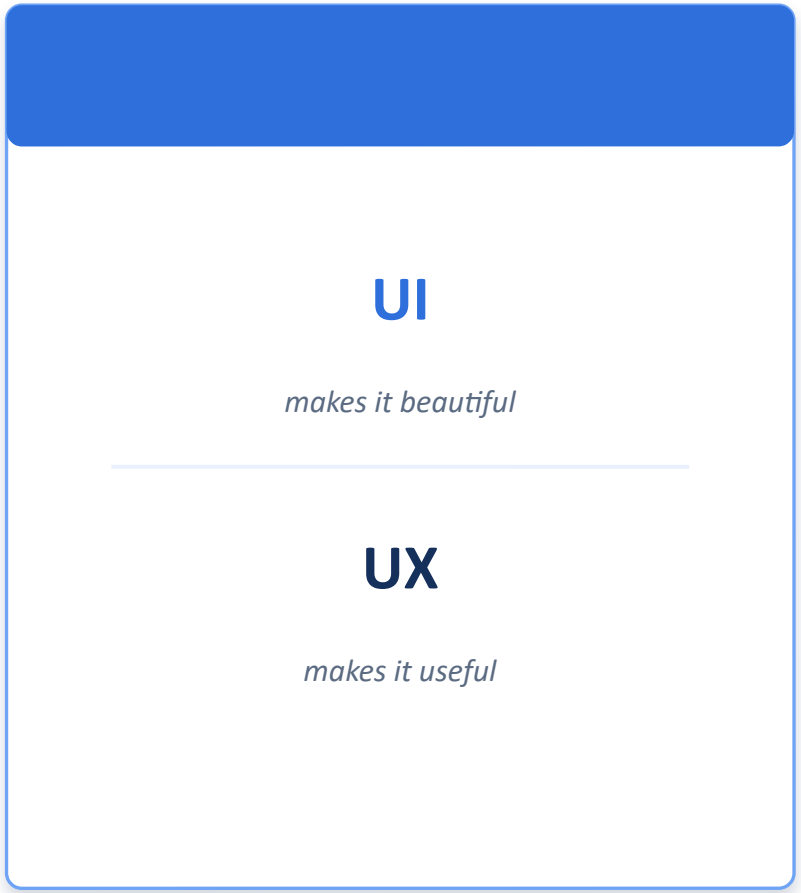
- Usability testing
- User research
- Personas

UI

- Layout
- Visual design
- Branding

Useful vs. Beautiful

- UX design is what makes an interface useful — it shapes how the product works.
- UI design is what makes an interface beautiful — it shapes how the product looks.
- Great products need both working together.



FRAMEWORKS

The Design Process

The FIVE Planes



1. Strategy

The foundation layer — why are we building this?

- **1. Strategy**
- 2. Scope
- 3. Structure
- 4. Skeleton
- 5. Surface

- What is the problem you are trying to solve?
- What are your user needs?
- How does your product fit within a business context (product objectives)?

2. Scope

Turning strategy into concrete requirements.

1. Strategy
2. **Scope**
3. Structure
4. Skeleton
5. Surface

- Defining functional specifications and content requirements of your platform.
- What are the features, and how might you prioritize them?
- The challenge is prioritizing features while minimizing tradeoffs.

3. Structure

Organizing information and user journeys.

1. Strategy
2. Scope
3. **Structure**
4. Skeleton
5. Surface

- Information architecture is concerned with how the information within an app is organized.
- User flows map out the specific journeys users embark on through the app to help solve their specific need.

4. Skeleton

Where design starts taking visible shape.

1. Strategy
2. Scope
3. Structure
4. **Skeleton**
5. Surface

- Achieve the structure of an interface design and information design.
- UX or Product Designers start designing, testing, and iterating on wireframes.

5. Surface

The sensory, visual finishing layer.

1. Strategy
2. Scope
3. Structure
4. Skeleton
5. **Surface**

- Communicate the brand, product, value and functionality in one cohesive image.
- UI Designers and Visual Designers start their work.
- Tools: Adobe XD, Sketch, Photoshop, Figma, etc.

BEST PRACTICES

UI Design Core Principles

What makes good UI?



CORE PRINCIPLE

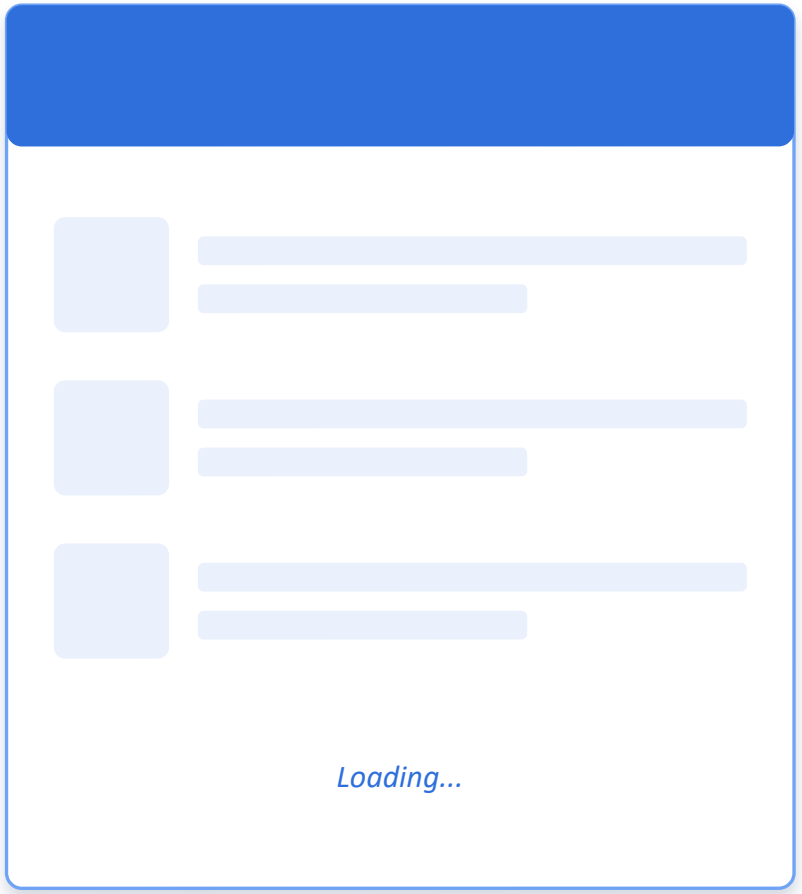
Clarity

- There's nothing worse than ambiguity in an app.
- Understand which design might help a user best achieve their goals.
- Let the user feel confident while navigating through your app.

CORE PRINCIPLE

Feedback

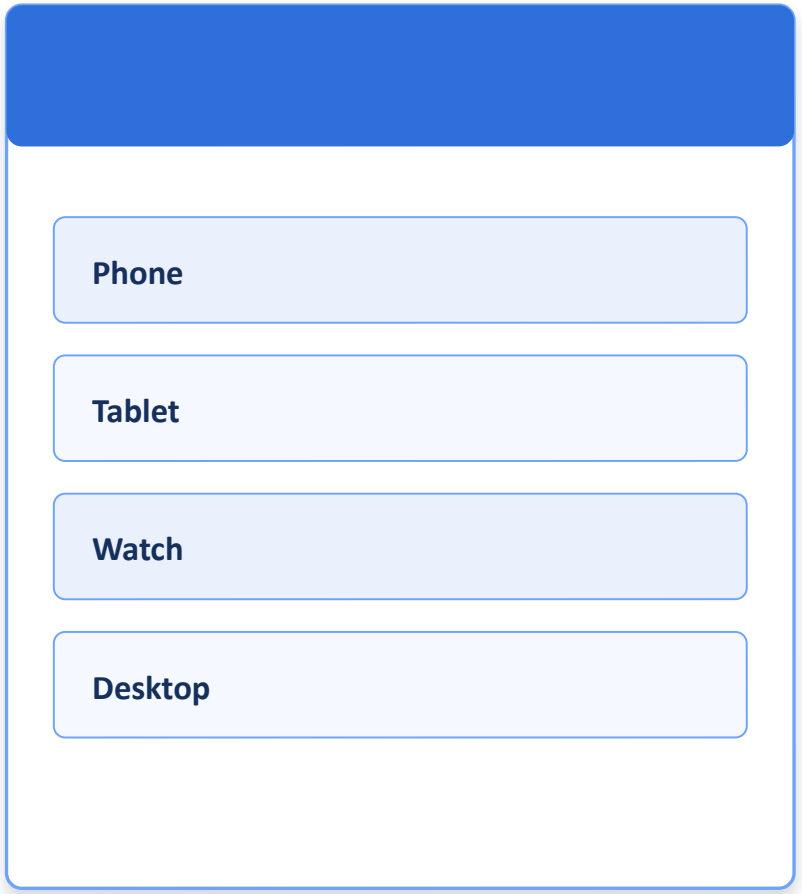
- Every action needs a reaction.
- This includes loading states, error and success messages, etc.
- Let the user know what is going on!



CORE PRINCIPLE

Consistency

- Be consistent in your UI elements.
- Once your users start learning how to use a product, they shouldn't have to relearn it.
- Good consistency and structure will make your users feel at home — across phone, tablet, watch and desktop.



CORE PRINCIPLE

Use Established Design Patterns

- Innovation is awesome and encouraged, but it shouldn't come at the expense of the user experience.
- Many established design principles are grounded in human perception.
- These are the best practices within the UI design space.

PATTERNS IN THE WILD

Familiar Patterns Build Trust

1

Lazy Registration

Let users try the product before asking them to sign up — reduce friction early on.

2

Card Layouts

Pinterest-style cards organize varied content into a scannable, consistent grid.

3

Autocomplete

Google's search suggestions help users complete their thought faster.

A CLOSER LOOK

Visual Hierarchy

How layout guides the eye

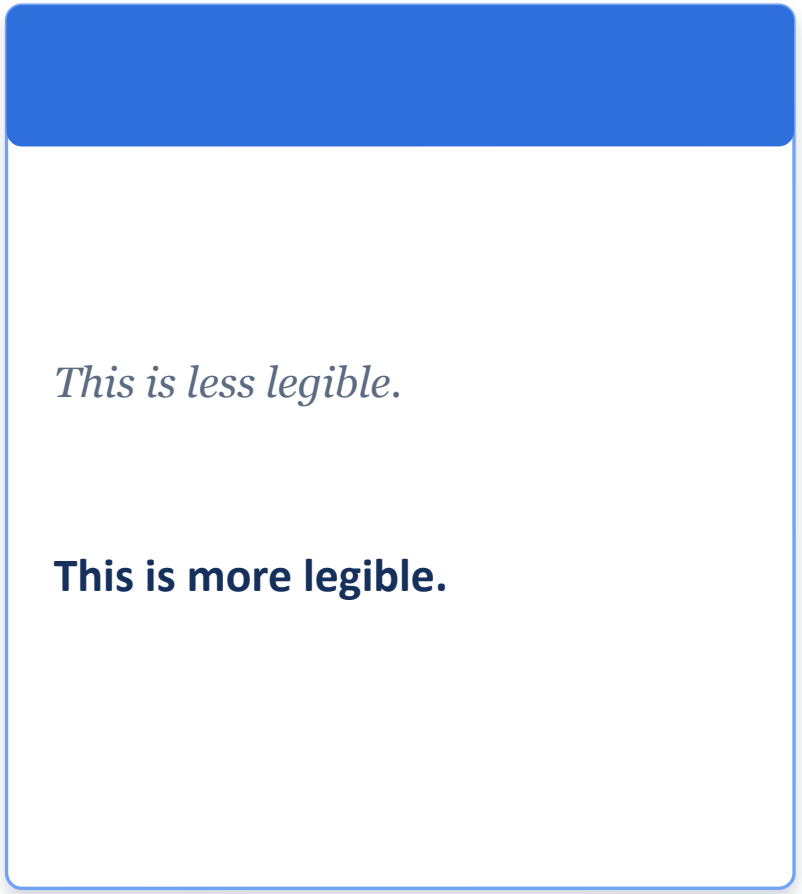


The Arrangement of Elements

- The arrangement of elements in a way that implies relative importance.
- The way we perceive information is affected by several factors that contribute to how we rank the hierarchy of content within the layout.
- Typography, white space and colors are the three main tools.

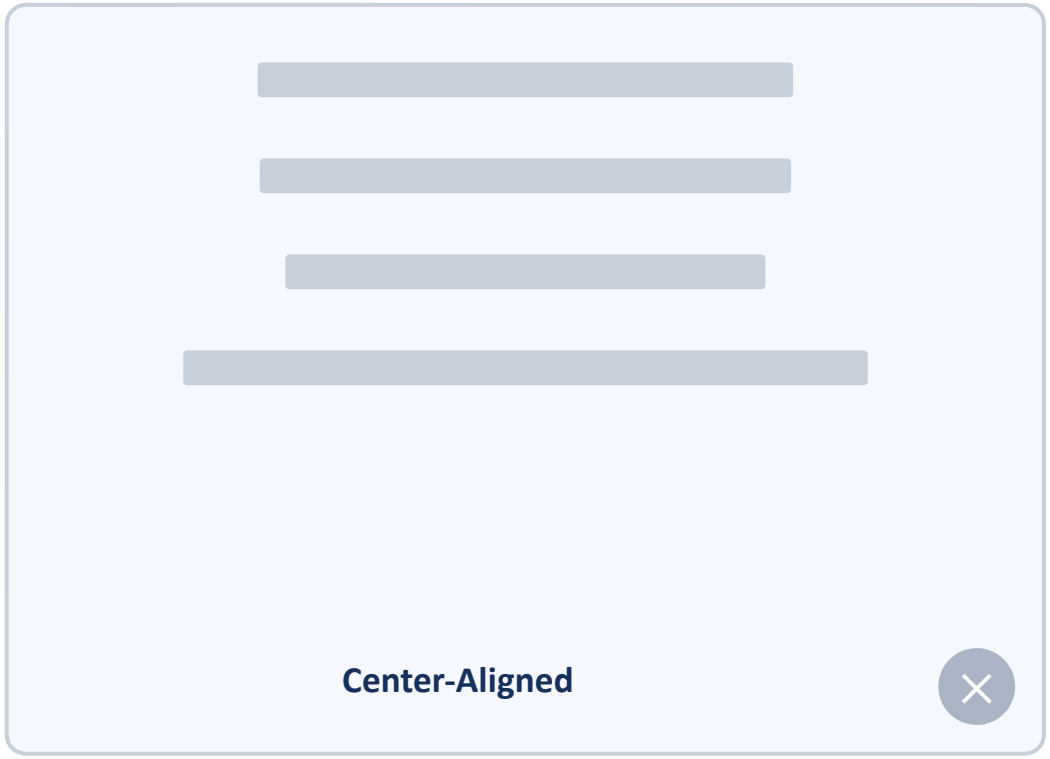
Legibility & Readability

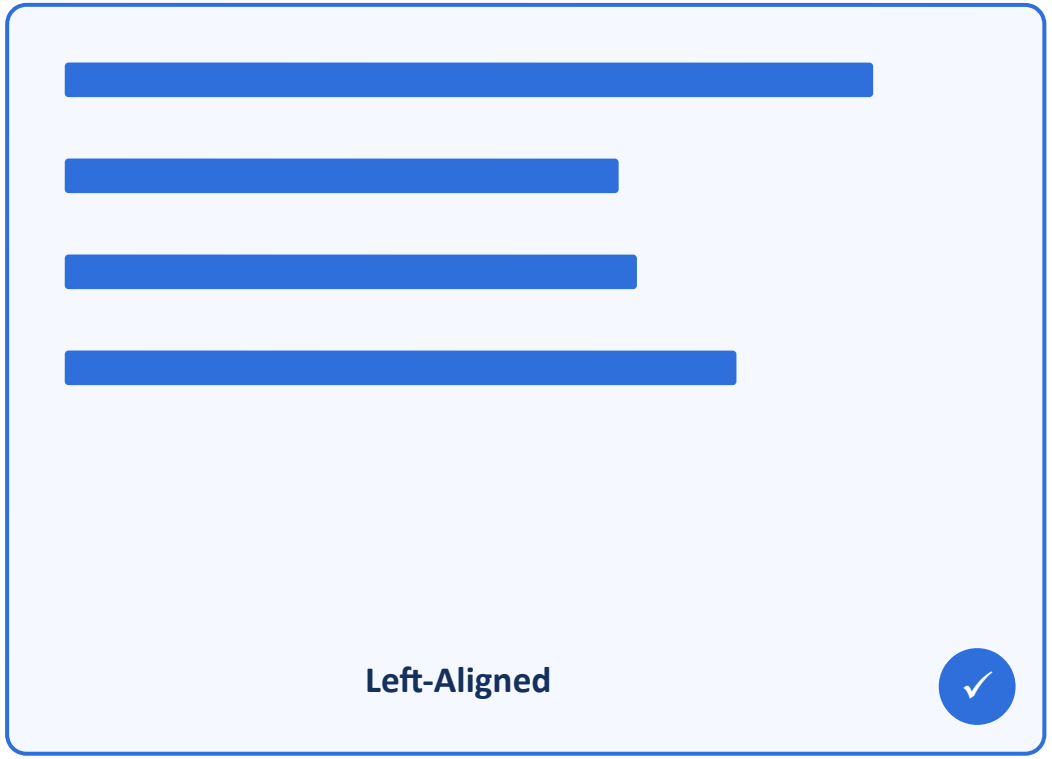
- The two primary factors that make good typography: legibility and readability.
- Legibility depends on the typeface — how each individual letterform is distinguishable from another.
- Readability is determined by how you manipulate a typeface to make it easier to comprehend.



Never Center Large Blocks of Copy

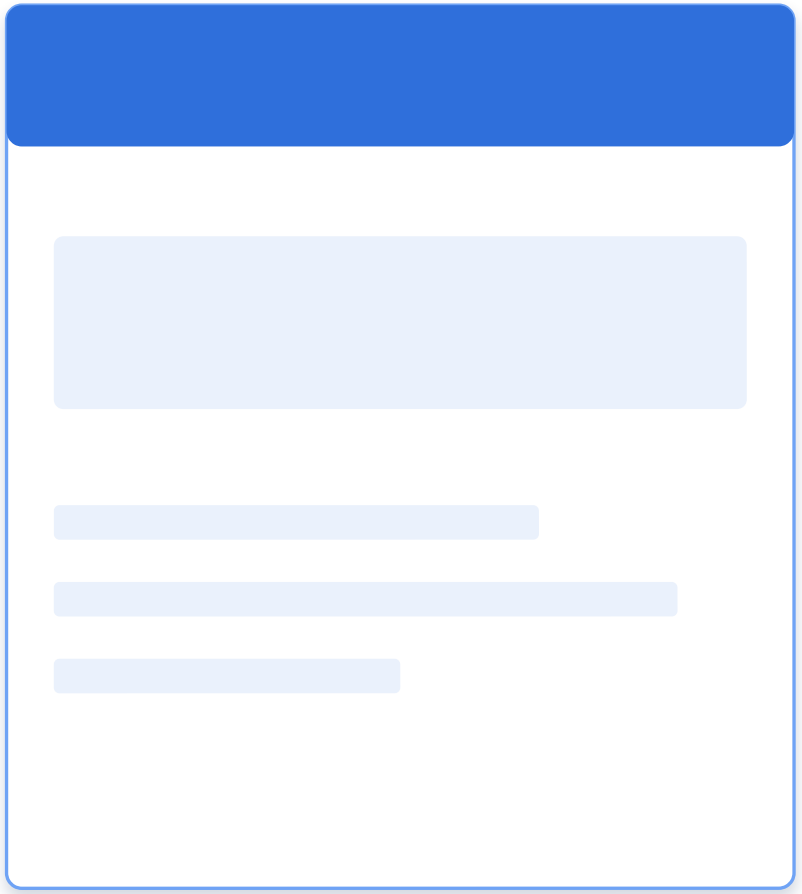
The eye knows exactly where the next line begins.





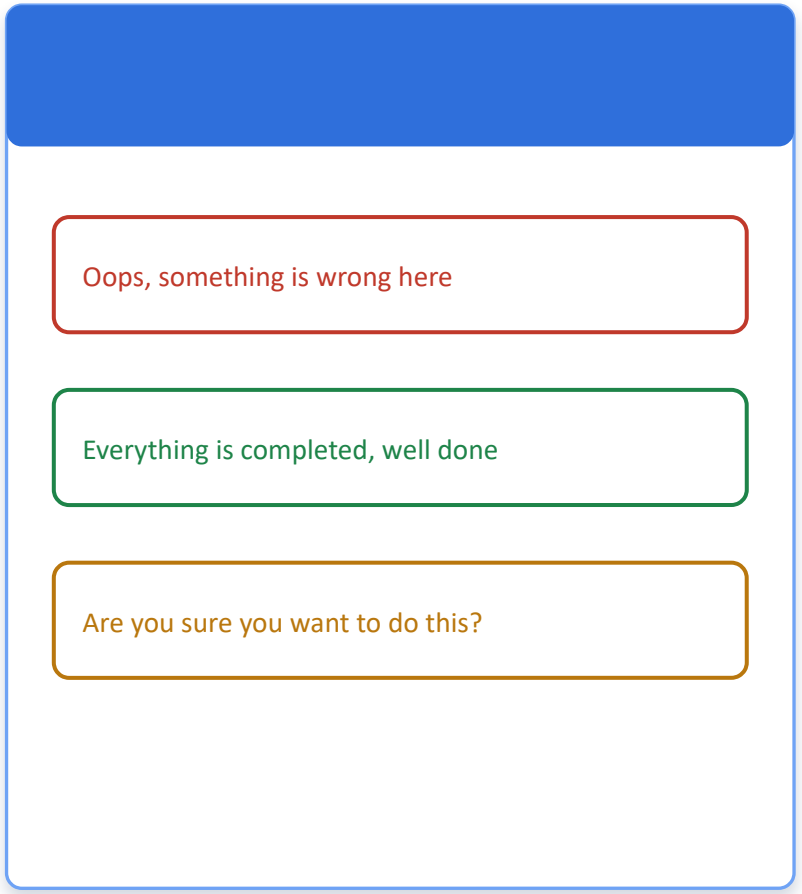
Give Content Room to Breathe

- White space helps with readability and comprehension immensely.
- Readers find it easier to focus on and process generously spaced content.
- White space can be used to create strong hierarchy — and line spacing directly affects comprehension.



Color Carries Meaning

- Colors have a key role in expressing the brand and intent.
- The more colors there are, the less impact each individual color will have.
- Keeping decent contrast between colors keeps a design accessible.



Patterns exist for a reason ;)

Use them wisely, and design with clarity, feedback, and consistency in mind.

FURTHER READING

References

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- InVision — Design Defined: User Interface Design
- InVision — Core Principles of UI Design
- UserTesting — UI vs UX
- Prototypr — Intro to User Interface Design
- Smashing Magazine — Comprehensive Guide to UX Design
- CareerFoundry — 5 Big Differences Between UX and UI Design
- Adobe Blog — 4 Golden Rules of UI Design
- Usability.gov — User Interface Elements
- InVision — Wireframe Examples

THANK YOU

